BRAC UNIVERSITY
Inspiring Excellence

Lecture 4
SOP & POS

	A	B	C	F
0	0	0	0	**1**
1	0	0	1	**1**
2	0	1	0	**0**
3	0	1	1	**0**
4	1	0	0	**1**
5	1	0	1	**1**
6	1	1	0	**0**
7	1	1	1	**0**

Min Terms : 0,1,4,5
[000,001,100,101]
$F=\Sigma(0,1,4,5)$

Max Terms : 2,3,6,7
[010,011,110,111]
$F=\Pi(2,3,6,7)$

CSE260 : Digital Logic Design

SOP & POS

Sum-of-Products (SOP) Expression:

A product term or a logical sum (OR) of several product terms.

Example: x+yz′, xy′+x′yz, AB+A′B’

Product-of-Sums (POS) Expression:

A sum term or a logical product (AND) of several sum terms.

Example: x(y+z′), (x+y′)(x′+y+z), (A+B)(A′+B’)

SOP & POS [Cont.]

Every boolean expression can either be expressed as sum-of-products or product-of-sums expression.

Example:

SOP: x'y + xy' + xyz

POS: (x + y')(x' + y)(x' + z')

MIN & MAX TERM

MIN & MAX TERM

Minterms are sum terms.

For Boolean functions, the minterms of a function are the terms for which the result is 1.

Boolean functions can be expressed as Sum- of-Minterms (SOM).

Maxterms are Product terms.

For Boolean functions, the maxterms of a function are the terms for which the result is 0.

Boolean functions can be expressed as Products-of-Maxterms (POM).

MIN & MAX TERM

	A	B	C	F
0	0	0	0	**1**
1	0	0	1	**1**
2	0	1	0	**0**
3	0	1	1	**0**
4	1	0	0	**1**
5	1	0	1	**1**
6	1	1	0	**0**
7	1	1	1	**0**

Min Terms : 0,1,4,5

[000,001,100,101] $F=\Sigma(0,1,4,5)$

Max Terms : 2,3,6,7

[010,011,110,111] $F=\Pi(2,3,6,7)$

MIN-SOP and MAX-POS

	A	B	C	F
0	0	0	0	**1**
1	0	0	1	**1**
2	0	1	0	**0**
3	0	1	1	**0**
4	1	0	0	**1**
5	1	0	1	**1**
6	1	1	0	**0**
7	1	1	1	**0**

Min Terms : 0,1,4,5 [000,001,100,101]

F=Σ(0,1,4,5)

F=A'B'C'+A'B'C+AB'C'+AB'C

Max Terms : 2,3,6,7 [010,011,110,111] F=Π(2,3,6,7)

F=(A+B'+C)(A+B'+C')(A'+B'+C) (A'+B'+C')

Min/SOP

0-Prime, 1-No prime

AND among literals

OR among terms

Max/POS

1-Prime, 0-No prime

OR among literals

AND among terms

Conversion of Function to SOP & POS

How to Convert into SOP

Steps

- Check if each term contains all variable,
  - if not then AND (x+x') if x is the missing term
- Remove the duplicates

How to Convert into SOP

Example:

F(A,B,C)=A+B'C

=A (B+B') (C+C') + B'C(A+A')

=(AB+AB') (C+C') + B'C(A+A')

=AB(C+C') + AB'(C+C') + B'C(A+A')

=ABC + ABC' + *AB'C* + AB'C' + *B'CA* + B'CA'

Now, removing duplicates and writing the literals in order,

=ABC + ABC' + AB'C + AB'C' + A'B'C

= 111, 110, 101, 100, 001

=Σ(1,4,5,6,7)

How to Convert into POS

Steps

1. Apply distributive law **(x+yz)=(x+y)(x+z)** if possible at every stage
2. If then terms, like x, are missing, append the term by OR **x.x'**
3. Check if any sum term is missing any variable, if so then OR missing variable. Again apply distributive law
4. Remove Duplicate terms

How to Convert into POS:

F(A,B,C) =A+B'C

= (A+B')(A+C)

= (A+B'+CC')(A+C+BB')

= (A+B'+C)(A+B'+C')(A+C+B)(A+C+B')

Now, removing duplicates and writing the literals in order,

= (A+B'+C)(A+B'+C')(A+B+C)

= 010, 011, 000

= π (2,3,0)

= π (0,2,3)

Example

Find SOP for F(w,x,y,z) =wy+x'z

Solution:

F(w,x,y,z)

= wy + x'z

= wy[(x+x')(z+z')] + x'z[(w+w')(y+y')]

= wy(xz+xz'+x'z+x'z') + x'z(wy+wy'+w'y+w'y')

= wyxz + wyxz' + wyx'z + wyx'z' + x'zwy + x'zwy' + x'zw'y + x'zw'y'

= wxyz + wxyz' + wx'yz + wx'yz' + wx'y'z + w'x'yz + w'x'y'z

= 1111,1110,1011,1010,1001,0011,0001

= Σ(15,14,11,10,9,3,1)

= Σ(1,3,9,10,11,14,15)

Example

Find POS for F(w,x,y,z) =wy+x'z

Solution:

F(w,x,y,z)

= wy+x'z

= (wy+x')(wy+z)

= (x'+w)(x'+y)(w+z)(y+z)

= (x'+w+yy')(x'+y+zz')(w+z+xx')(y+z+xx')

= (x'+w+y)(x'+w+y')(x'+y+z)(x'+y+z')(w+z+x)(w+z+x')(y+z+x)(y+z+x')

= (x'+w+y+zz')(x'+w+y'+zz')(x'+y+z+ww')(x'+y+z'+ww')
(w+z+x+yy')(w+z+x'+yy')(y+z+x+ww')(y+z+x'+ww')

= (w+x'+y+z)(w+x'+y+z')(w+x'+y'+z)(w+x'+y'+z')
(w+x'+y+z)(w'+x'+y+z)(w+x'+y+z')(w'+x'+y+z')
(w+x+y+z)(w+x+y'+z)(w+x'+y+z)(w+x'+y'+z)
(w+x+y+z)(w'+x+y+z)(w+x'+y+z)(w'+x'+y+z)

= (w+x'+y+z)(w+x'+y+z')(w+x'+y'+z)(w+x'+y'+z')(w'+x'+y+z)
(w'+x'+y+z')(w+x+y+z)(w+x+y'+z)(w'+x+y+z)

= 0100,0101,0110,0111,1100,1101,0000,0010,1000

= Π(4,5,6,7,12,13,0,2,8) = Π(0,2,4,5,6,7,8,12,13)

Example

Find POS for F(w,x,y,z) =wy+x'z

Alternate Solution: Same thing can be done by taking all missing terms together

F(w,x,y,z)
= wy+x'z
= (wy+x')(wy+z)
= (x'+w)(x'+y)(w+z)(y+z)
= (x'+w+yy'+zz')(x'+y+zz'+ww')(w+z+xx'+yy')(y+z+xx'+ww')
= (x'+w+y+zz')(x'+w+y'+zz')(x'+y+z+ww')(x'+y+z'+ww')
(w+z+x+yy')(w+z+x'+yy')(y+z+x+ww')(y+z+x'+ww')
= (w+x'+y+z)(w+x'+y+z')(w+x'+y'+z)(w+x'+y'+z')
(w+x'+y+z)(w'+x'+y+z)(w+x'+y+z')(w'+x'+y+z')
(w+x+y+z)(w+x+y'+z)(w+x'+y+z)(w+x'+y'+z)
(w+x+y+z)(w'+x+y+z)(w+x'+y+z)(w'+x'+y+z)
= (w+x'+y+z)(w+x'+y+z')(w+x'+y'+z)(w+x'+y'+z')(w'+x'+y+z)
(w'+x'+y+z')(w+x+y+z)(w+x+y'+z)(w'+x+y+z)
= 0100,0101,0110,0111,1100,1101,0000,0010,1000
= Π(4,5,6,7,12,13,0,2,8) = Π(0,2,4,5,6,7,8,12,13)

Example

Convert the following function to its corresponding canonical **POS** form using boolean algebraic formulas: F(a,b,c,d) = (a+b+c'd')(b+d)

F(a,b,c,d)

= (a+b+c'd')(b+d)

= (a+b+c')(a+b+d')(b+d)

= (a+b+c'+d')(a+b+c'+d)(a+b+d'+c')(a+b+d'+c)(b+d+a')(b+d+a)

= (a+b+c'+d')(a+b+c'+d)(a+b+d'+c')(a+b+d'+c)(b+d+a'+c')(b+d+a'+c)(b+d+a+c')(b+d+a+c)

= (a+b+c'+d')(a+b+c'+d)(a+b+c'+d')(a+b+c+d')(a'+b+c'+d)(a'+b+c+d)(a+b+c'+d)(a+b+c+d)

= (a+b+c'+d')(a+b+c'+d)(a+b+c+d')(a'+b+c'+d)(a'+b+c+d)(a+b+c+d)

= ∏ (3, 2, 1, 10, 8, 0)

= ∏ (0, 1, 2, 3, 8, 10)

Example

Convert the following function to its corresponding canonical POS form using distributive law:
F(A, B, C, D) = ABC′+ AC′ + D

F(A, B, C, D) = ABC′+ AC′ + D
= AC’(B+1)+D
= AC’+D
= (A+D)(C′+D)
= (A+D)(C′+D+AA′)
= (A+D)(A+C′+D)(A′+C′+D)
= (A+D+BB′)(A+C′+D+BB′)(A′+C′+D+BB′)
= (A+B+D)(A+B′+D)(A+B+C′+D)(A+B′+C′+D)(A′+B+C′+D)(A′+B′+C′+D)
= (A+B+D+CC′)(A+B′+D+CC′)(A+B+C′+D)(A+B′+C′+D)(A′+B+C′+D)(A′+B′+C′+D)
= (A+B+C+D)(A+B+C′+D)(A+B′+C+D)(A+B′+C′+D)(A+B+C′+D)(A+B′+C′+D)(A′+B+C′+D)(A′+B′+C′+D)

Now, removing duplicates and writing the literals in order:
= (A+B+C+D)(A+B+C′+D)(A+B′+C+D)(A+B′+C′+D)(A′+B+C′+D)(A′+B′+C′+D)
= 0000, 0010, 0100, 0110, 1010, 1110
= π (0, 2, 4, 6, 10, 14)

NB: You see, for such examples, the original equation is simplified and then the rest of the POS operation took place. So YES, if needed, the question given can be simplified before POS. However, if you are thinking whether this can be done without simplifying, YES, it can be done. However, the process will be longer. Check next slide.

Example

Convert the following function to its corresponding canonical POS form using distributive law:
F(A, B, C, D) = ABC′+ AC′ + D

F(A,B,C,D) = ABC′+ AC′ + D
= (A+D)(A+C′+D)(B+A+D)(B+C′+D)(C′+A+D)(C′+D)
= (A+D)(A+C′+D)(B+A+D)(B+C′+D+AA′)(C′+A+D)(C′+D+AA′)
= (A+D)(A+C′+D)(A+B+D)(A+B+C′+D)(A′+B+C′+D)(A+C′+D)(A+C′+D)(A′+C′+D)
= (A+D+BB′)(A+C′+D+BB′)(A+B+D)(A+B+C′+D)(A′+B+C′+D)(A+C′+D+BB′)(A+C′+D+BB′)(A′+C′+D+BB′)
= (A+B+D)(A+B′+D)(A+B+C′+D)(A+B′+C′+D)(A+B+D)(A+B+C′+D)(A′+B+C′+D)(A+B+C′+D)(A+B′+C′+D)(A+B+C′+D)(A+B′+C′+D)
(A′+B+C′+D)(A′+B′+C′+D)
= (A+B+D+CC′)(A+B′+D+CC′)(A+B+C′+D)(A+B′+C′+D)(A+B+D+CC′)(A+B+C′+D)(A′+B+C′+D)(A+B+C′+D)(A+B′+C′+D)(A+B+C′+D)
(A+B′+C′+D)(A′+B+C′+D)(A′+B′+C′+D)
= (A+B+C+D)(A+B+C′+D)(A+B′+C+D)(A+B′+C′+D)(A+B+C′+D)(A+B′+C′+D)(A+B+C+D)(A+B+C′+D)(A+B+C′+D)(A′+B+C′+D)(A+B+C′+D)
(A+B′+C′+D)(A+B+C′+D)(A+B′+C′+D)(A′+B+C′+D)(A′+B′+C′+D)

Now, removing duplicates and writing the literals in order:
= (A+B+C+D)(A+B+C′+D)(A+B′+C+D)(A+B′+C′+D)(A′+B+C′+D)(A′+B′+C′+D)
= 0000, 0010, 0100, 0110, 1010, 1110
= π (0, 2, 4, 6, 10, 14)
NB: This is if you don’t simplify the given function, still the same result. Just a longer process.